

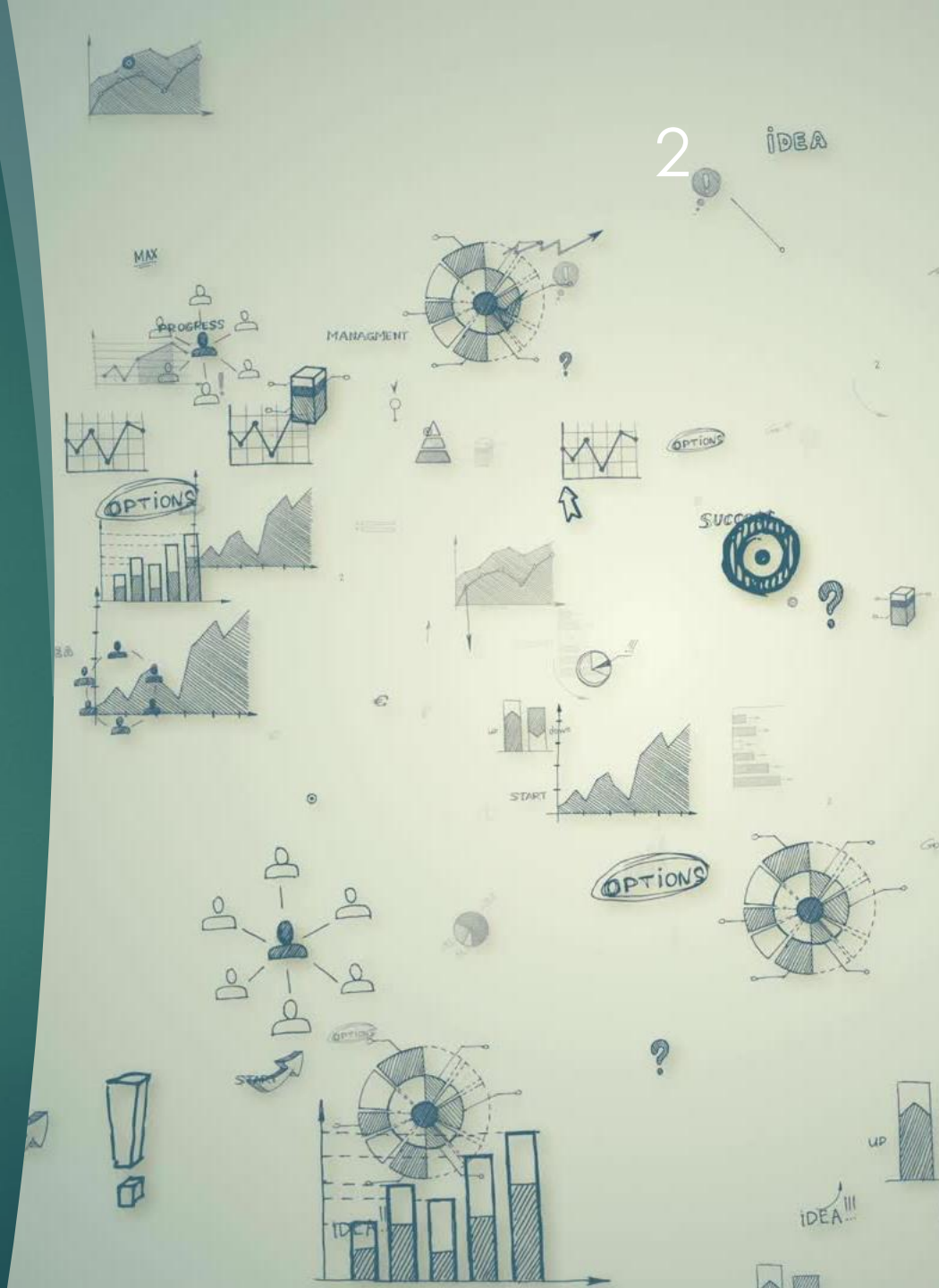
Bayesian decision theory

CHAPTER 3



Chapter 1 Roadmap

- Introduction
- The maximum a posteriori classifier
- Bayesian classification
- The naive Bayes classifier
- The minimax classifier
- Normally distributed features
- Arbitrarily distributed features
- Gaussian mixtures



Introduction

- Bayes Decision Theory is a fundamental statistical approach to the problem of pattern classification.
- It is based on the assumption that the decision problem is posed in **probabilistic terms**.
- All relevant probability values are assumed to be known.
- In practice, these probabilities are estimated from available data.
- The goal is to design a classifier that minimizes the average risk (cost) of misclassification.

Introduction

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- This chapter develops the theoretical foundation for optimal statistical classifiers, establishing the probabilistic framework upon which most pattern recognition methods are built.
- **CORE IDEA**
 - Given a **feature vector** $\mathbf{x}$, **assign** it to the **class** ω_i that **minimizes** the **expected cost** of **misclassification**.
- **WHY IT MATTERS**
 - Bayes classifiers achieve the **theoretically lowest possible error rate** — the *Bayes error* — serving as the **gold standard benchmark**.

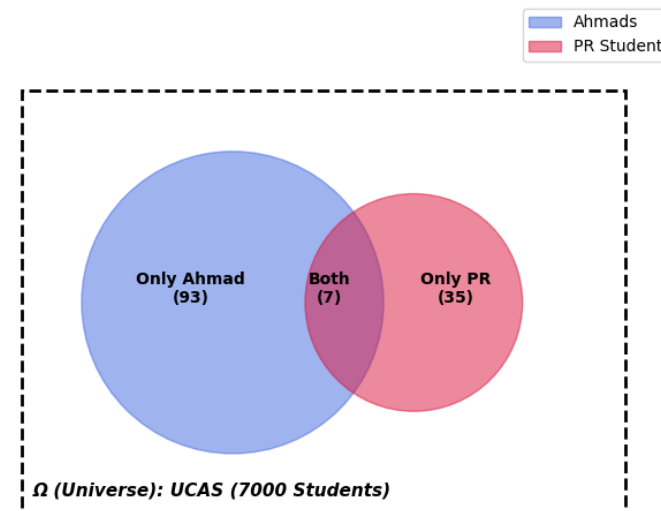
PROBLEM SETUP

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- Suppose we have 7000 student in UCAS, 100 are named “Ahmad”
- And in “Pattern Recognition” Course, we have 7 students named “Ahmad” out of 42 total students.
- $P(\text{Ahmad}, \text{PR}) = P(\text{Ahmad} | \text{PR}) * P(\text{PR}) = P(\text{PR} | \text{Ahmad}) * P(\text{Ahmad})$



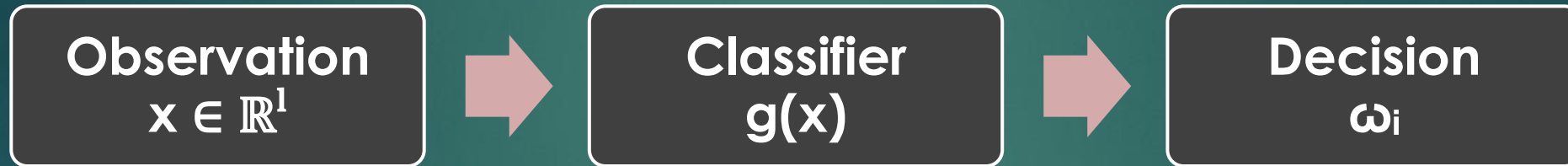
UCAS Student Distribution: Bayesian Sample Space



PROBLEM SETUP

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- We observe a feature vector $\mathbf{x} \in \mathbb{R}^1$ drawn from an unknown class ω_i , $i = 1, \dots, M$. The goal is to design a classifier that maps $\mathbf{x} \rightarrow \omega_i$ with minimum error.



STATISTICAL QUANTITIES

- $P(\omega_i)$ — prior probability of class ω_i
- $p(\mathbf{x} \mid \omega_i)$ — class-conditional PDF
- $P(\omega_i \mid \mathbf{x})$ — posterior probability
- $p(\mathbf{x})$ — unconditional (evidence) PDF

M CLASSES SETTING

- Classes: $\omega_1, \omega_2, \dots, \omega_M$
- Prior constraint:
- $\sum_i P(\omega_i) = 1$
- Priors encode knowledge about class frequencies before observing $\mathbf{x}$.

BAYES DECISION THEORY

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- **Bayes' Theorem in Classification**

- Bayes' Theorem — posterior = likelihood × prior / evidence

- $P(\omega_i | x) = p(x | \omega_i) \cdot P(\omega_i) / p(x)$

1. **Total probability** (evidence / normalization constant)

- $p(x) = \sum_i p(x | \omega_i) \cdot P(\omega_i)$

2. **PRIOR $P(\Omega_i)$**

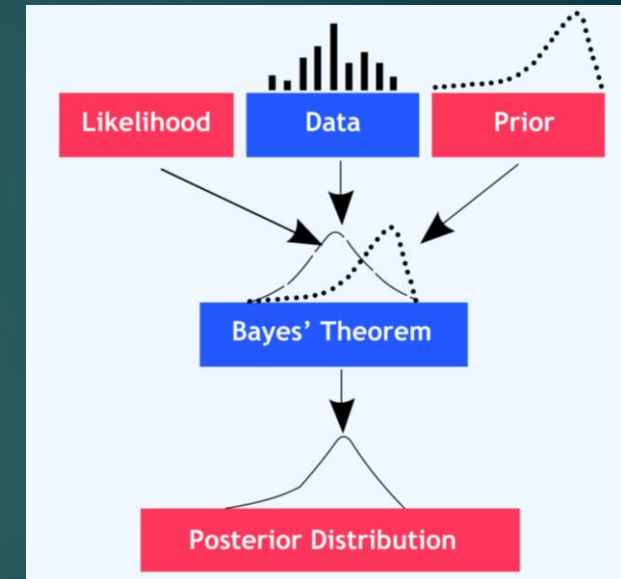
- **Probability** that a randomly chosen pattern belongs to **class ω_i** before observing any feature.

3. **LIKELIHOOD $P(X | \Omega_i)$**

- The **probability density of feature x** given that the **true class is ω_i** . Characterizes each class's distribution.

4. **POSTERIOR $P(\Omega_i | X)$**

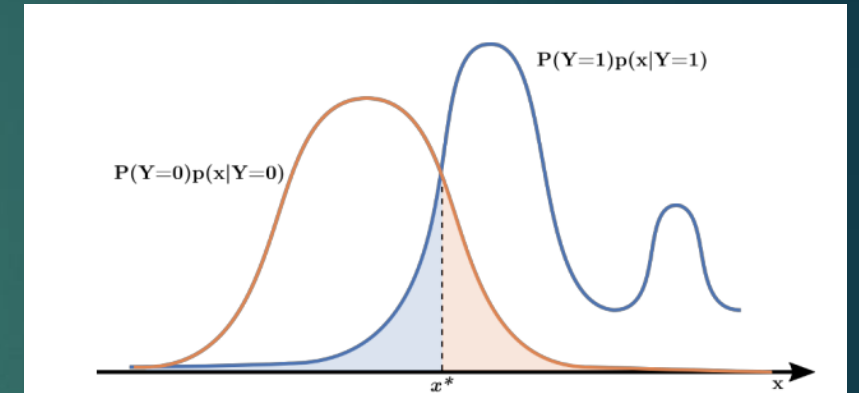
- Updated belief about class membership *after* observing x . This is what the Bayes classifier maximises.



The Bayes Classifier (Minimum Error)

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- *Decide the class with highest posterior probability*
- **BAYES DECISION RULE**
- Assign x to class ω_k if $P(\omega_k | x) \geq P(\omega_i | x)$ for all $i \neq k$.
- Maximum A Posteriori (MAP) decision rule
 - decide $\omega_k \Leftrightarrow k = \operatorname{argmax}_i P(\omega_i | x)$



EQUIVALENT FORM

- Since $p(x)$ is constant for all classes, MAP is equivalent to:
- decide $\omega_k \Leftrightarrow k = \operatorname{argmax}_i [p(x | \omega_i) \cdot P(\omega_i)]$
- No need to compute the normalization $p(x)$.

SPECIAL CASE: EQUAL PRIORS

- When $P(\omega_1) = P(\omega_2) = \dots = P(\omega_m) = 1/M$
- decide $\omega_k \Leftrightarrow k = \operatorname{argmax}_i p(x | \omega_i)$
- Maximum Likelihood (ML) rule.

Probability of Error & Bayes Error

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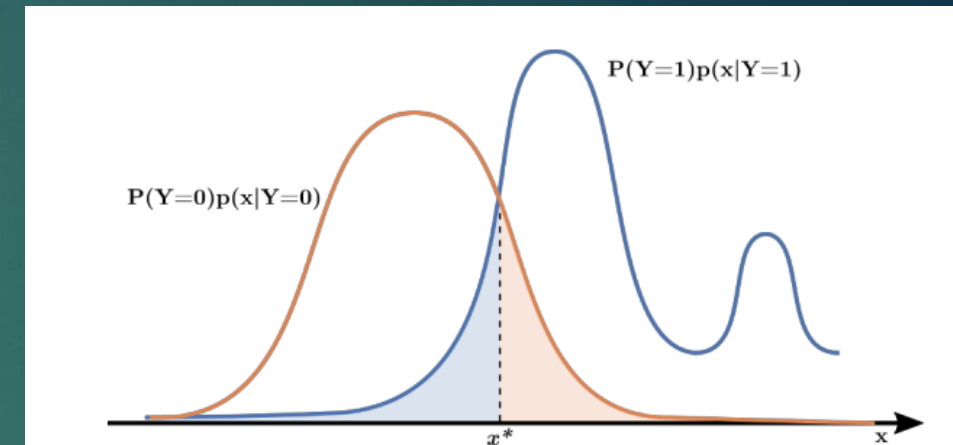
- *The theoretical lower bound on classification error*
- Conditional probability of error given observation x
 - $P(\text{error} \mid x) = 1 - \max_i P(\omega_i \mid x)$

- **Bayes error rate**

- $P^*(\text{error}) = \int [1 - \max_i P(\omega_i \mid x)] p(x) dx$

- **KEY INSIGHT**

- The Bayes classifier minimizes the average probability of error.
 - **No deterministic classifier can achieve a lower error rate than $P^*(\text{error})$ for the same underlying probability distributions. This quantity serves as the fundamental benchmark.**



TWO-CLASS CASE ($M = 2$)

- $P(\text{error} \mid x) = \min[P(\omega_1 \mid x), P(\omega_2 \mid x)]$
- Decide ω_1 if $P(\omega_1 \mid x) > P(\omega_2 \mid x)$, else ω_2 .

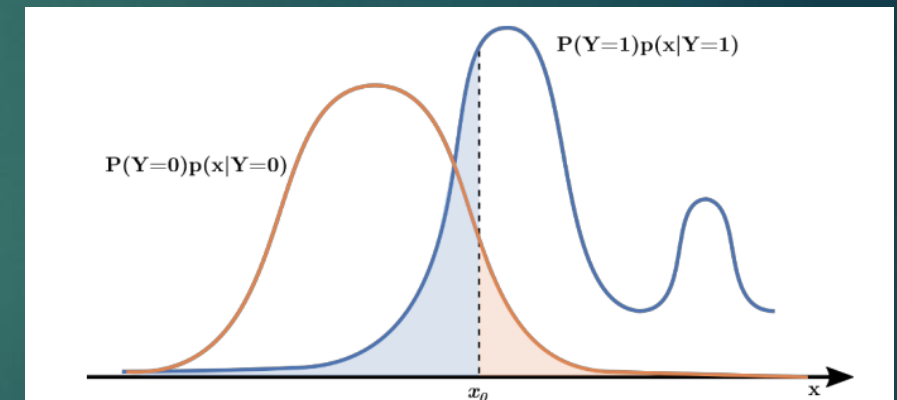
OVERLAP REGIONS

- Bayes error is non-zero whenever class-conditional distributions overlap. The more overlap, the higher $P^*(\text{error})$.

Loss Functions & Risk

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- *Incorporating asymmetric costs of misclassification*
- LOSS (COST) MATRIX
- $\lambda(a_i | \omega_j)$ is the loss incurred when we take action a_i (decide class ω_i) but the true class is ω_j . Typically $\lambda(a_i | \omega_i) = 0$ (correct) and $\lambda(a_i | \omega_j) > 0$ for $i \neq j$.
- Conditional risk of action a_i given observation x
 - $R(a_i | x) = \sum_j \lambda(a_i | \omega_j) \cdot P(\omega_j | x)$
- Minimum risk (Bayes) decision rule
 - decide $a_k \Leftrightarrow k = \operatorname{argmin}_i R(a_i | x)$



- TRUE = ω_1 , decide ω_2
- $\lambda(a_2 | \omega_1)$
- FALSE NEGATIVE COST

- TRUE = ω_2 , decide ω_1
- $\lambda(a_1 | \omega_2)$
- FALSE POSITIVE COST

The Loss Matrix Layout

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- In a 2-class problem (like Bit 0 vs. Bit 1, or Healthy vs. Sick), the matrix is structured as follows:

Action / True Class	$\omega 1$ (True Class 1)	$\omega 2$ (True Class 2)
α_1 (Decide Class 1)	λ_{11}	λ_{12}
α_2 (Decide Class 2)	λ_{21}	λ_{22}

- λ_{12} (the False Negative / Type II Error):
- You Decided α_1 (E.G., "Normal") But The Truth Was $\omega 2$ (E.G., "Attack")
- This Is Often The Highest Cost In Security Or Medical Systems.

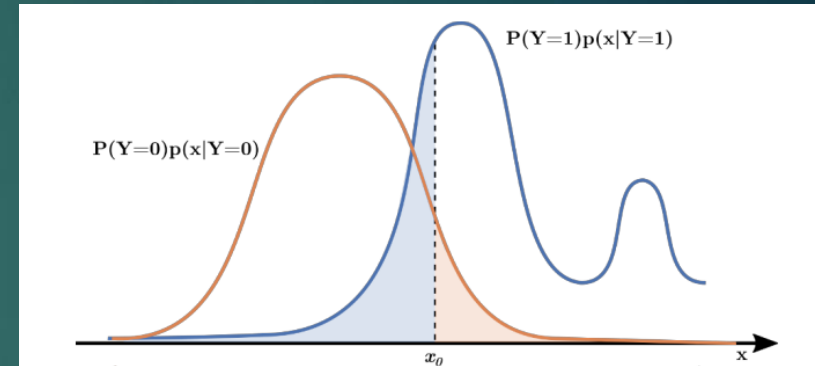
- λ_{21} (the False Positive / Type I Error):
- You Decided α_2 (E.G., "Attack") But The Truth Was $\omega 1$ (E.G., "Normal").
- This Results In A "False Alarm." It Has A Cost (Wasted Time), But Usually Less Than λ_{12}

Medical Diagnosis (Cancer vs. Healthy)

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1. The Scenario

- Imagine a patient undergoes a scan (x).
- Based on the "fuzziness" of the image, the AI calculates the following posterior probabilities:
 - $P(\text{Healthy} | x) = 0.90$ (90% chance they are fine)
 - $P(\text{Cancer} | x) = 0.10$ (10% chance they have cancer)



2. The Loss Matrix $\lambda(\alpha_i | \omega_i)$

- In a standard MAP rule, you would choose "Healthy."
- But in medicine, the costs of being wrong are asymmetric (uneven).

Action / True Class	ω_0 : Healthy	ω_1 : Cancer
α_1 Decide Healthy	0 (Correct)	1000 (Missed diagnosis - Fatal)
α_2 Decide Cancer	10 (False alarm - Stress/Biopsy)	0 (Correct)

Medical Diagnosis (Cancer vs. Healthy)

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3. Calculate Conditional Risk ($R(\alpha_i|x)$)

- We calculate the "expected pain" for each decision.

- Risk of deciding "Healthy" (α_1)

$$\begin{aligned} R(\alpha_1|x) &= \lambda(\alpha_1|\omega_1) * P(\omega_1|x) + \lambda(\alpha_1|\omega_2) * P(\omega_2|x) \\ &= 0 * 0.9 + 1000 * 0.1 = 100 \end{aligned}$$

- Risk of deciding "Cancer" (α_2)

$$\begin{aligned} R(\alpha_2|x) &= \lambda(\alpha_2|\omega_1) * P(\omega_1|x) + \lambda(\alpha_2|\omega_2) * P(\omega_2|x) \\ &= 10 * 0.9 + 0 * 0.1 = 9 \end{aligned}$$

Medical Diagnosis (Cancer vs. Healthy)

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4. The Bayes Decision Rule

- We choose the action that minimizes risk:

$$\operatorname{argmin}\{R(\alpha_1|x), R(\alpha_2|x)\} = \operatorname{argmin}\{100, 9\} = \alpha_2$$

- Decision: The system decides "**Cancer**", even though there is only a 10% probability.
- Summary
 - **MAP (Maximum Probability)**: Would have said "Healthy" (90% probability).
 - **Bayes Risk (Minimum Cost)**: Says "Cancer" because the **penalty** for **being wrong** about a **healthy person** (10) is **much lower than** the **penalty** for being **wrong about a sick person** (1000).

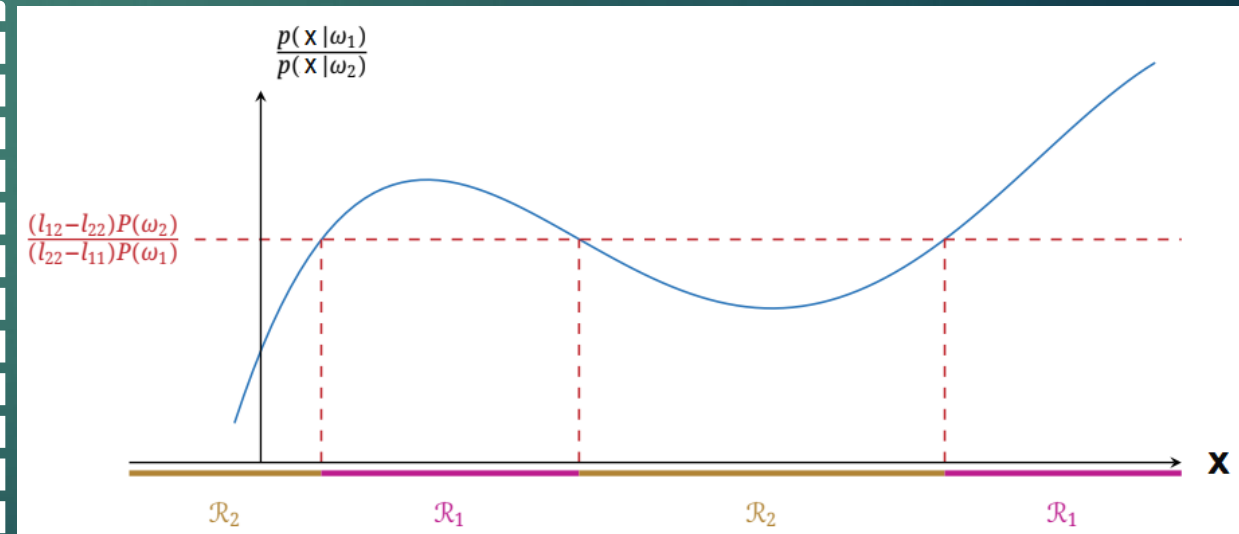
Likelihood Ratio

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- Likelihood Ratio Test ($C = 2$)
 - Decide ω_1 if the likelihood ratio $\Lambda(x)$ exceeds threshold θ ; otherwise decide ω_2 .
- Likelihood ratio test — decide ω_1 ; where $\lambda_{ij} = \lambda(a_i | \omega_j)$
 - $\Lambda(x) = p(x | \omega_1) / p(x | \omega_2) > \theta \rightarrow \frac{[\lambda_{12} - \lambda_{22}] \cdot P(\omega_2)}{[\lambda_{21} - \lambda_{11}] \cdot P(\omega_1)} > \theta$

DECISION REGIONS

- $\mathcal{R}_1$: region of $\mathbb{R}^1$ assigned to ω_1
- $\mathcal{R}_2$: region of $\mathbb{R}^1$ assigned to ω_2
- Decision boundary separates $\mathcal{R}_1$ and $\mathcal{R}_2$
- Boundary: set where $\Lambda(x) = \theta$



MAP Decoding in Digital Communications

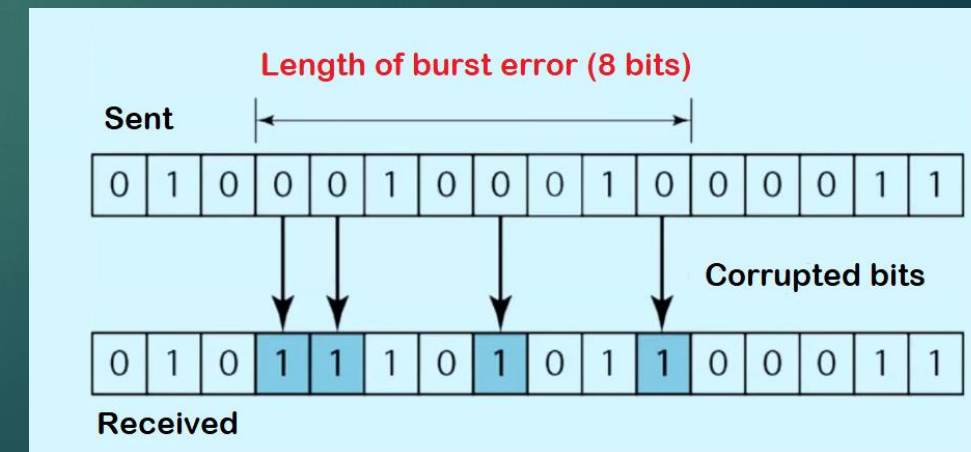
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- A Case Study in Bayesian Pattern Recognition

1. The Physical Signal Model

- In a Wi-Fi transmission, bits are represented by voltage levels. Due to walls and interference, the receiver gets a signal corrupted by **Additive White Gaussian Noise (AWGN)**.

- **Target Class ω_0 (Bit 0):** Ideal voltage = **-5V**
 - **Target Class ω_1 (Bit 1):** Ideal voltage = **+5V**
- **Observed Data (x):** The router measures **+1V**



MAP Decoding in Digital Communications

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2. Likelihood vs. Prior

- **Likelihood** $p(x|\omega_i)$
 - The sensor sees (+1V). This is physically closer to the (+5V) "Bit 1" signature.
 - *Pure evidence suggests: **Bit 1***
- **The Prior** $p(\omega_i)$
 - The protocol context (e.g., a silent VoIP call) indicates that "0" bits are extremely common right now.
 - $p(\omega_0) = 0.9$ & $p(\omega_1) = 0.1$

MAP Decoding in Digital Communications

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3. The MAP Decision Rule

- The Maximum A Posteriori (MAP) classifier calculates the probability of the class *given* the measured voltage:

$$P(\omega_i|x) \propto p(x|(\omega_i)) * p(\omega_i)$$

- Even though the **likelihood** of (+1V) is higher for Bit 1, the **high prior probability of Bit 0** (0.9) acts as a weight that pulls the decision toward class (ω_0)

MAP Decoding in Digital Communications

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4. The Result: Tie-Breaking with Intelligence

- When the signal is "fuzzy" or ambiguous (like +1 V), the router doesn't just guess based on the closest voltage.
- It uses its "expectation" to break the tie.
- **Decision:** The router decodes this as **Bit 0**.
- **Impact:**
 - This minimizes the **Bit Error Rate (BER)**.
 - In high-interference environments, relying on context (MAP) is more reliable than relying solely on raw, noisy measurements (Maximum Likelihood).

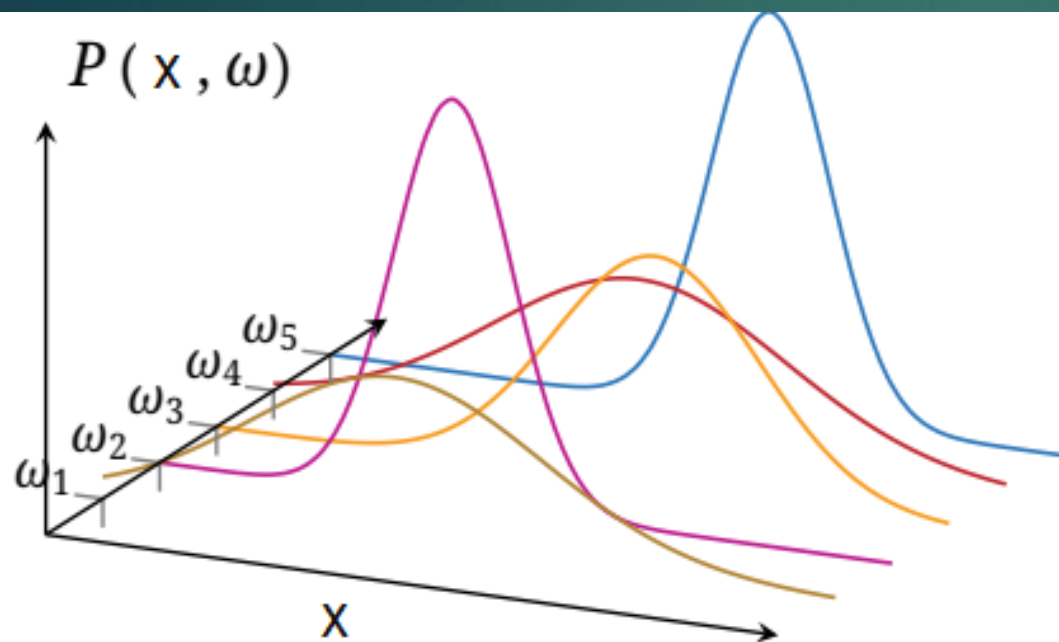
MAP Decoding in Digital Communications

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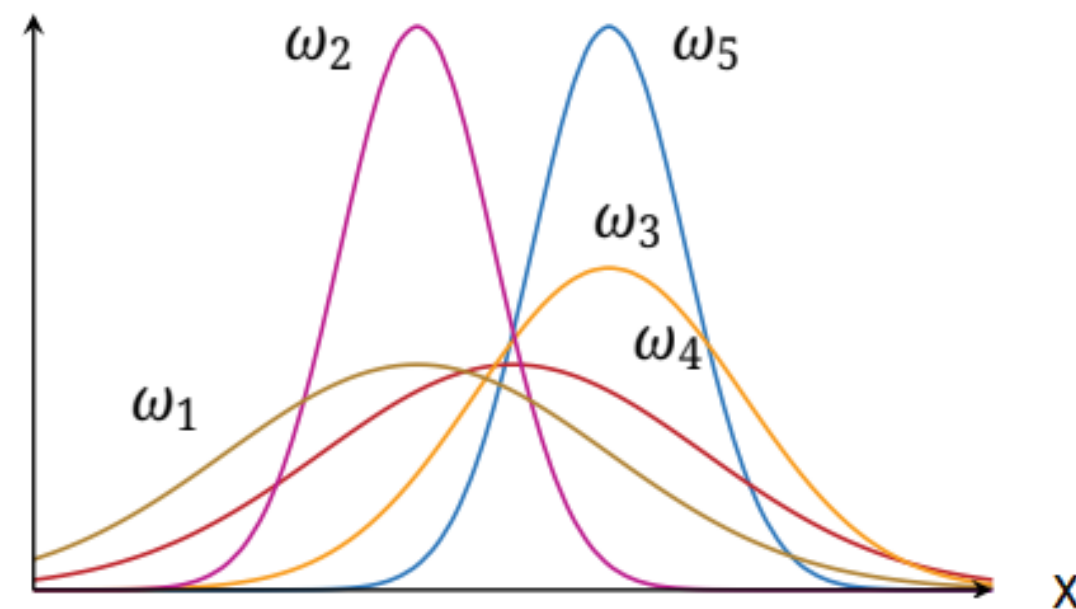
- **IEEE Wi-Fi standards that rely on MAP receivers**
 - All modern Wi-Fi standards from 802.11n onward use **soft-decision decoding** which is mathematically equivalent to MAP decoding of bit sequences.
- **IEEE 802.11n**
 - Introduced **Low-Density Parity-Check (LDPC) codes** as an option. LDPC decoders use belief propagation – a form of MAP decoding (max-product algorithm).
- **Your phone's modem:**
 - Every time you stream a video, the baseband processor computes LLRs (MAP per bit) and feeds them to an LDPC or Viterbi decoder.
 - This is Bayesian decision theory running at gigabit speeds.

More than two classes

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(a) Full joint distribution



(b) Marginal distributions

Example of Optimal Decision Boundary

SECTION 3.3.2

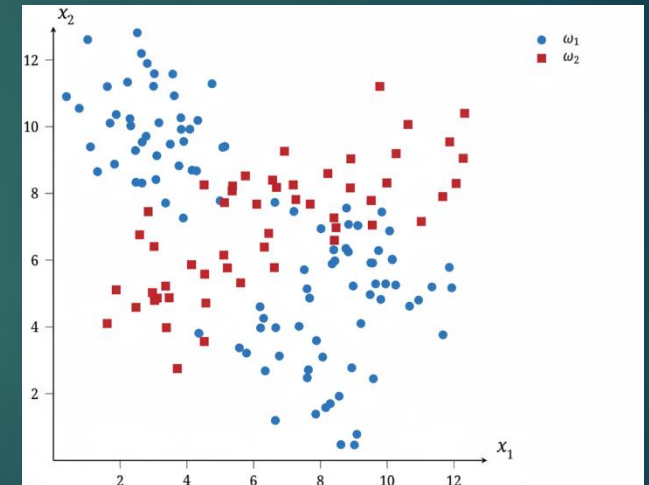
Example of Optimal Decision Boundary

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- In Bayesian classification, we consider feature vectors $X = (x_1, x_2)$.
- In this model, we use a mixture of seven isotropic Gaussian densities for each class.
- **The Likelihood Function**

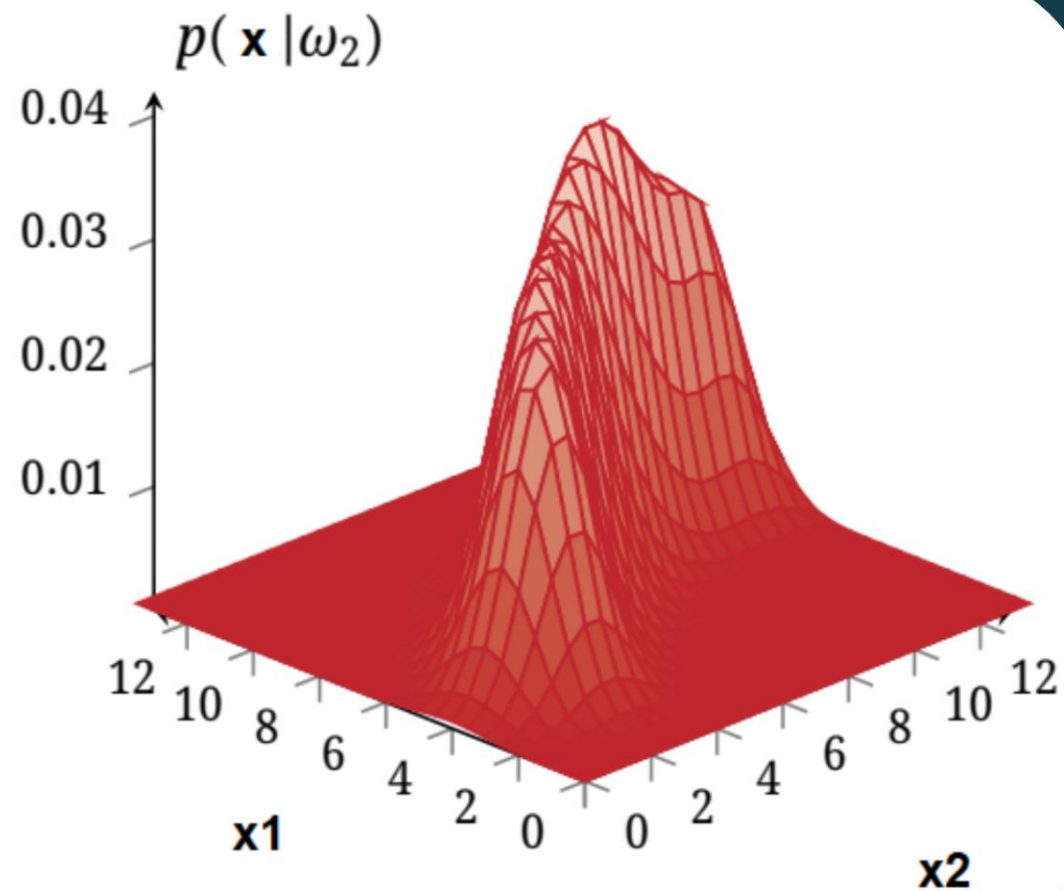
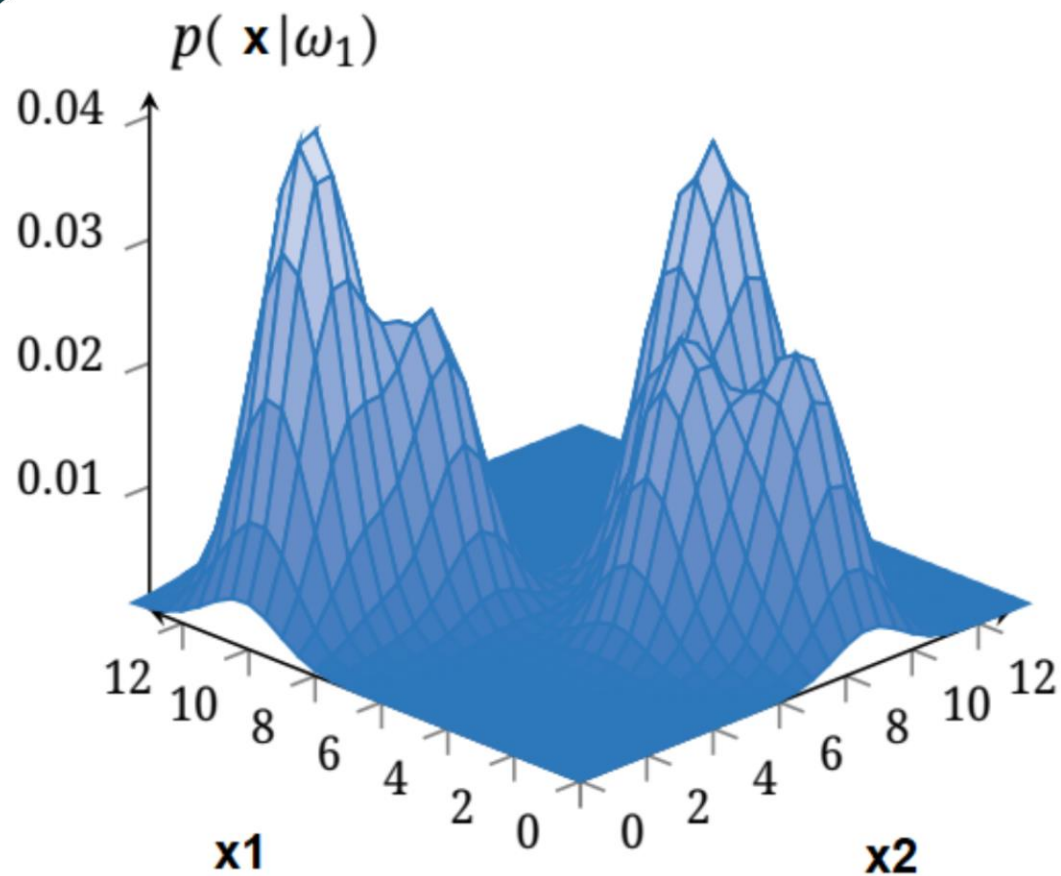
$$p(x|\omega_j) = \frac{1}{7} \sum_{i=1}^7 N(x; \mu_{ij}, \sigma_j^2)$$

- Where we define $\sigma_1 = \sigma_2 := 1$.
- Both classes are equally likely, meaning the prior probabilities are $P(\omega_j) = 0.5$.



Likelihood Visualization

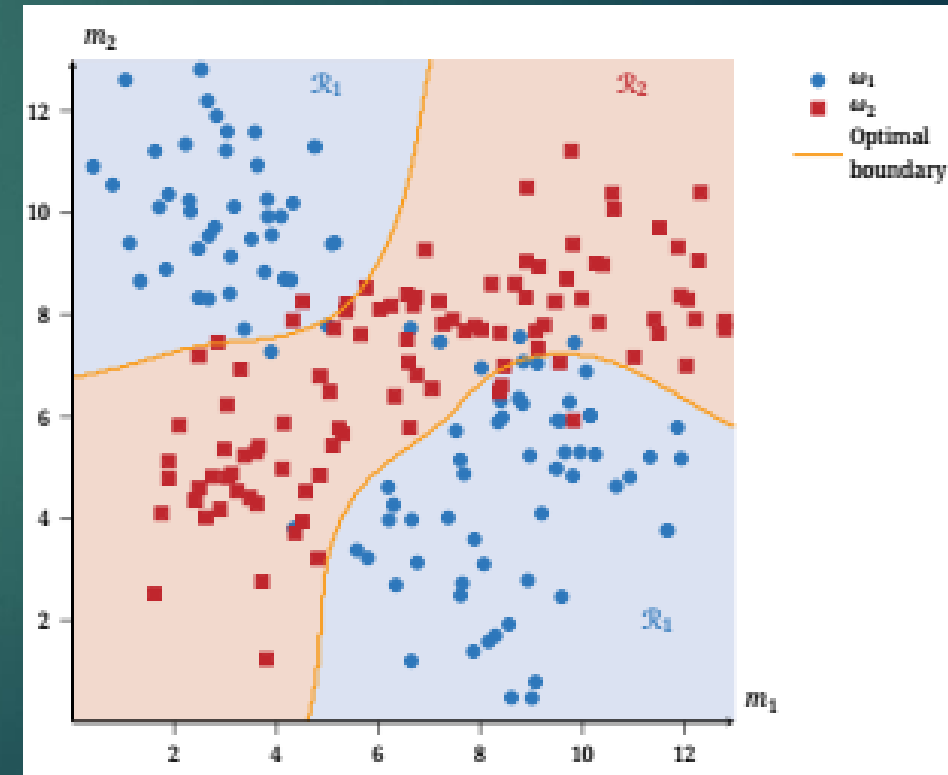
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Optimal Decision Boundary

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- The boundary corresponds to the set of points where the posterior probabilities are equal:
- $P(\omega_1 | x) = P(\omega_2 | x)$
- **Region $\mathcal{R}_1$:** Points where $P(\omega_1 | x) > P(\omega_2 | x)$
- **Region $\mathcal{R}_2$:** Points where $P(\omega_2 | x) > P(\omega_1 | x)$
- This creates the orange "Optimal Boundary" seen in the classification plots, separating the blue circles from the red squares.



Classification Error Analysis

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- There is a fundamental difference between the mathematical limit of the model and the results of a specific trial.

Type of Error	Value	Description
Theoretical Error (P_e)	$\approx 6.16\%$	Calculated using the known underlying distributions
Empirical Error	8.00%	The actual errors made on a specific dataset D (16 out of 200 samples).

- The Key Lesson**
 - The empirical error deviates from the theoretical error because it is subject to **random perturbations**.
 - It depends entirely on the specific dataset sampled.
- "More reliable estimates of the true probability of error can be obtained by using a larger test sample."*

The naive Bayes classifier

Simplifying the curse of Dimensionality through
conditional independence

SECTION 3.3.3

The Naive Bayes assumption

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- In real-world tasks, estimating full joint distribution $p(x_1, \dots, x_d | \omega)$ requires vast amounts of data — exponential in "d".
- The **naive Bayes assumption** treats features as conditionally independent given the class.

$$p(x_1, \dots, x_d | \omega) = \prod_{i=1}^d p(x_i | \omega)$$

- **Parameter reduction (categorical case):** from $2^d * C$ to $2 * d * C$ parameters → dramatically less training data required.
- For binary feature vector of length "10", full distribution requires 1024 parameters to estimate, while Naïve Bayes Requires only 20 parameters

The Naive Bayes assumption

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- MAP decision rule simplifies to:

$$\hat{\omega} = \operatorname{argmax}_{\omega_j} p(\omega_j) \prod_{i=1}^d p(x_i | \omega_j)$$

- Despite the often unrealistic independence assumption, the classifier works remarkably well on many real datasets (e.g., text classification, medical diagnosis).
- Naive Bayes performs poorly when features are strongly correlated — as we'll see on the reference example (decision boundary limitations).

The Naive Bayes

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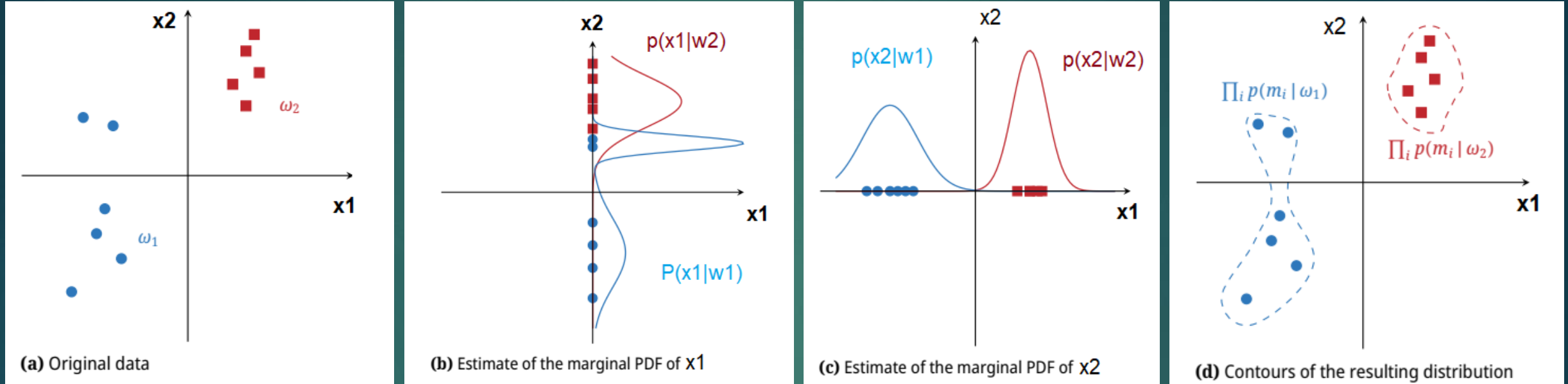
- When considering continuous features, the underlying class conditional distribution for each feature $p(x_i|\omega_j)$ often is assumed to be a Gaussian

$$p(x|\omega_j) = \prod_{i=1}^d \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(x_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

- This additional assumption leads to the naming “Gaussian naive Bayes”
- The overall distribution $p(x|\omega) = N(x; \mu_\omega, \Sigma_\omega)$ follows a Gaussian with a diagonal covariance matrix Σ_ω

The Naive Bayes

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- For Naive Bayes we also can use Bernoulli distributions or multinomial variants
- In general, it is also possible to utilize various distributions within a single naive Bayes classifier.

Naive Bayes Real-World Application

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Application: Apache Lucene

- Automated Document Categorization in Lucene & Elasticsearch
 - The Apache Lucene core library includes a **ClassificationModule** with a **NaiveBayesClassifier**.
 - It's used by platforms like **Elasticsearch**, **Solr**, and **LinkedIn** to categorize incoming news feeds, job postings, or support tickets.
 - Training on 1 million articles takes minutes; classification of a new article takes <10 ms.
- **Feature Extraction**
 - An article is converted into a "**Bag of Words**."
 - The classifier focuses on *word frequencies*.
 - Words like "Ceasefire" or "drones" are high-frequency features for the "News" class.

Naive Bayes Real-World Application

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Application: Apache Lucene

- **The Multinomial Mathematical Model**

- The system calculates the probability of a category (C) given the words (w) in the document:

$$p(C | w_1, w_2, \dots, w_n) \propto \prod_{i=1}^d p(C | w_i)$$

- **Why Lucene Uses It:**

- **Scale:** Handles 100,000+ distinct words (features) without crashing.
- **Speed:** Multiplications are computationally "cheap," allowing for **real-time indexing**.
- **Accuracy:** For long documents, the sheer volume of words compensates for the "naive" independence assumption.

The Minimax classifier

Decision making under uncertain Prior probability

SECTION 3.3.4

The Minimax Classifier

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- What if we don't know the true prior distribution?
- Classical Bayesian classifiers rely on either:
 - The a posteriori distribution $P(\omega \mid \mathbf{x})$
 - The a priori distribution $P(\omega)$
- Knowing the a posteriori is very hard in real-world applications.
- Even estimating the a priori requires expert domain knowledge — and that estimate may be wrong.

Key Question

*What happens when we **design** a classifier with the **wrong a priori** probability?*

The minimax classifier is designed to remain robust even when the true prior is unknown.

The Minimax Classifier — Risk Analysis

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- The Risk is a Linear Function of the Prior $P(\omega_1)$

$$R = \underbrace{\lambda_{22} + (\lambda_{12} - \lambda_{22}) \int_{\mathcal{R}_1} p(x|\omega_2)dx}_{R_{\minimax}} + P(\omega_1) \underbrace{\left[(\lambda_{11} - \lambda_{22}) + (\lambda_{21} - \lambda_{11}) \int_{\mathcal{R}_2} p(x|\omega_1)dx - (\lambda_{12} - \lambda_{22}) \int_{\mathcal{R}_1} p(x|\omega_2)dx \right]}_{=0}$$

Linear relationship

Risk R is linear in $P(\omega_1)$.
The decision regions $\mathcal{R}_i$ are the design parameters controlling ordinate and slope.

Design mismatch

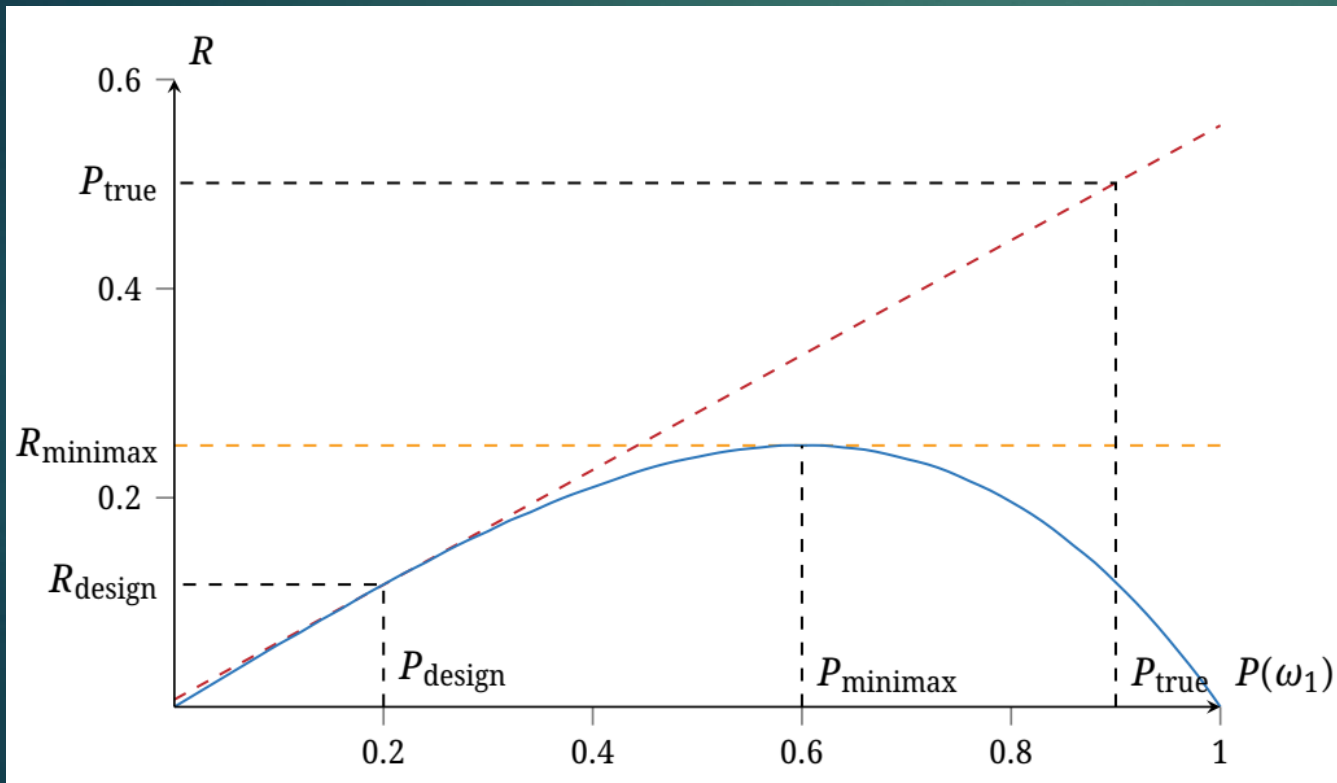
If the true prior $P_{\text{true}} \neq P_{\text{design}}$, the actual risk lies on a different tangent — possibly much higher.

Minimax solution

Set slope = 0 via the decision regions.
This makes R independent of $P(\omega_1)$, achieving constant risk $R_{\minimax}$.

Visualizing Risk

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Blue curve

Optimal Bayesian risk at each $P(\omega_1)$.
Zero at endpoints, peaks near P_{minimax} .

Yellow line

Constant minimax risk R_{minimax} — the horizontal tangent at the peak.

Red tangent

Actual risk when classifier is designed for P_{design} but deployed at P_{true} .

The Minimax Strategy

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1 Set slope = 0

- 1 Choose decision regions R_i such that the slope equals zero.
This eliminates dependency on $P(\omega_1)$.

2 Find worst-case prior

- 2 The chosen regions implicitly correspond to the optimal Bayesian classifier for P_{minimax} — the prior that maximizes risk.

3 Achieve constant risk

- 3 The result is constant $R = R_{\text{minimax}}$ regardless of the actual prior.
The tangent line is horizontal.

4 Trade-off

- 4 In classical pattern recognition, the **optimal Bayesian classifier often outperforms minimax** — but **only** if $P_{\text{design}} \approx P_{\text{true}}$.

Origin: The minimax approach comes from **game theory** — **minimizing the worst-case** outcome against an adversarial opponent.

Minimax Classifier-Real-World Example

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- **APPLICATION: RADAR THREAT DETECTION**
- **The Challenge: The Unknown Enemy**
 - In radar defense, we must distinguish between **Birds/Noise (ω_0)** and **Enemy Aircraft (ω_1)**.
 - The problem? We have no way of knowing the *A Priori* probability of an attack.
 - It could be 0.0001% today and 50% tomorrow.

▪ **The Bayes Problem**

- A standard Bayes classifier relies on $p(\omega_1)$
- If we assume attack probability is low, the system might ignore a real threat to avoid false alarms.

▪ **The Minimax Solution**

- We design a boundary that stays **fixed** regardless of the priors.
- We assume the "worst-case" prior—the one that maximizes our total risk.

Minimax Classifier-Real-World Example

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- **The "Equal Risk" Principle**

- The Minimax classifier finds a decision threshold where the probability of error is **constant**, regardless of how often an aircraft appears. $R(\omega_0) = R(\omega_1)$

- **Numerical intuition**

- Suppose a radar has two possible prior probabilities for “threat”: 0.001 (peace time) or 0.5 (wartime).
- A MAP classifier tuned for peacetime would miss 50% of threats during war.
- A minimax detector sets its threshold so that the **maximum risk (cost of false alarm + cost of miss) is equal** for both extremes.
- The result is a single robust decision rule that performs **gracefully under all conditions** – a cornerstone of modern radar and sonar systems.

Normally Distributed Features

SECTION 3.3.5

Why Normal Distributions?

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- **Central Limit Theorem**

- **If a feature is generated** by summing **many independent** contributions, it is reasonable to **approximate** its **distribution** with a **Gaussian** — even when individual components are not normally distributed.

- **Mathematical Convenience**

- Closed-form solutions for decision boundaries exist under Gaussian assumptions.

- **Ubiquity in Nature**

- Many real-world measurements (height, signal noise, sensor readings) are approximately Gaussian.

- **Tractable Multivariate Extension**

- The multivariate Gaussian generalizes cleanly to d-dimensional feature spaces.

Univariate Normal Distribution

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- A random variable “x” is normally (or Gaussian) distributed with expectation μ and variance σ^2 if its distribution can be described by the density:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

X	The random variable (observed feature value)
μ (mu)	Expectation — the mean / centre of the distribution
σ (sigma)	Standard deviation — controls the spread/width
σ^2	Variance — square of standard deviation

Multivariate Normal Distribution

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- Definition

- A random vector $x \in \mathbb{R}^d$ is multivariate Gaussian with mean vector $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$:

$$p(x) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} \times e^{(-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu))}$$

- For the 2D case, the covariance matrix decomposes as:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_2\sigma_1 & \sigma_2^2 \end{bmatrix}$$

Key Symbols

x	Feature vector (d-dimensional)
d	Number of features (dimensions)
μ	Mean vector (class centre)
Σ	Covariance matrix (shape & orientation)
$ \Sigma $	Determinant of Σ
Σ^{-1}	Inverse of covariance matrix
ρ	Pearson correlation coefficient

Multivariate Normal Distribution- MAP Classifier

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- MAP Rule (Discriminant function)

$$P(\omega_1|x) > P(\omega_2|x) \quad \rightarrow \quad p(x|(\omega_1)) * p(\omega_1) > p(x|(\omega_2)) * p(\omega_2)$$

$$\ln (p(x|(\omega_1))) + \ln (p(\omega_1)) > \ln (p(x|(\omega_2))) + \ln (p(\omega_2))$$

$$\ln (p(x|(\omega_1))) + \ln (p(\omega_1)) > \ln (p(x|(\omega_2))) + \ln (p(\omega_2))$$

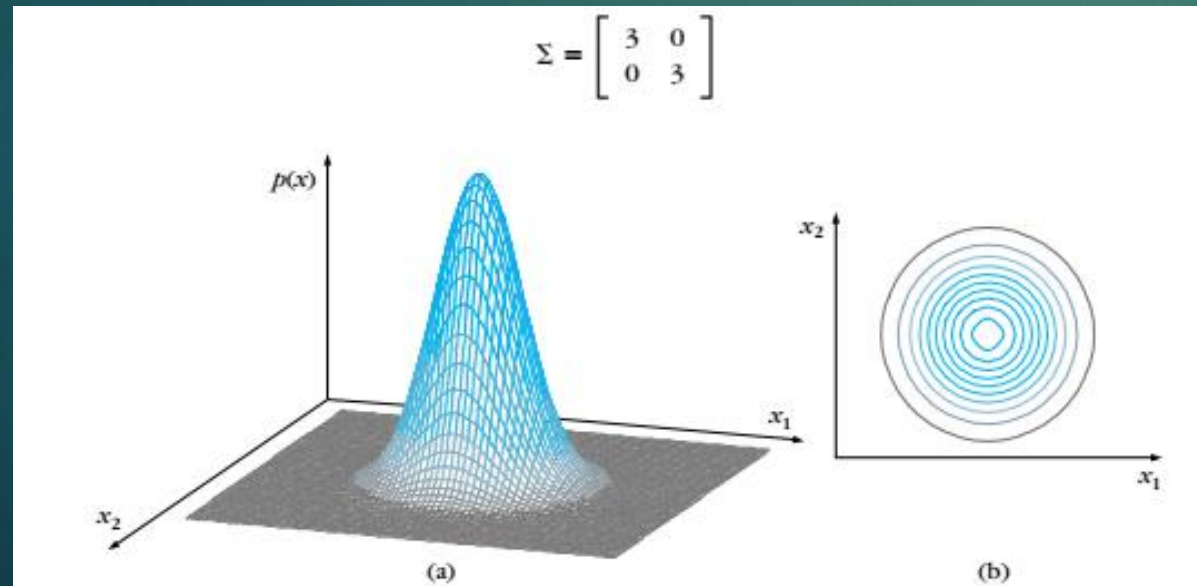
$$\ln (p(x|(\omega_1))) + \ln (p(\omega_1)) = -\frac{1}{2} (x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_1| + \ln(p(\omega_1))$$

Multivariate Normal Distribution- MAP Classifier

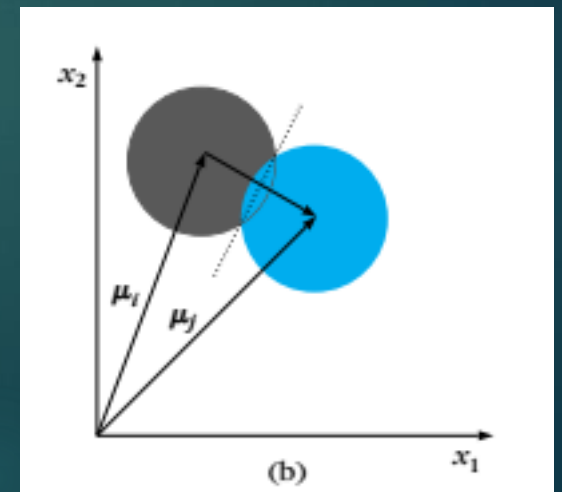
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- Case 1 $\Sigma_i = \sigma^2 I$ (equal, spherical)

$$-\frac{1}{2} (x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_i| + \ln(p(\omega_i)) = \frac{1}{2\sigma^2} \|x - \mu_i\|^2 + \ln P(\omega_i)$$



- Boundaries $\perp$ to connection lines between means



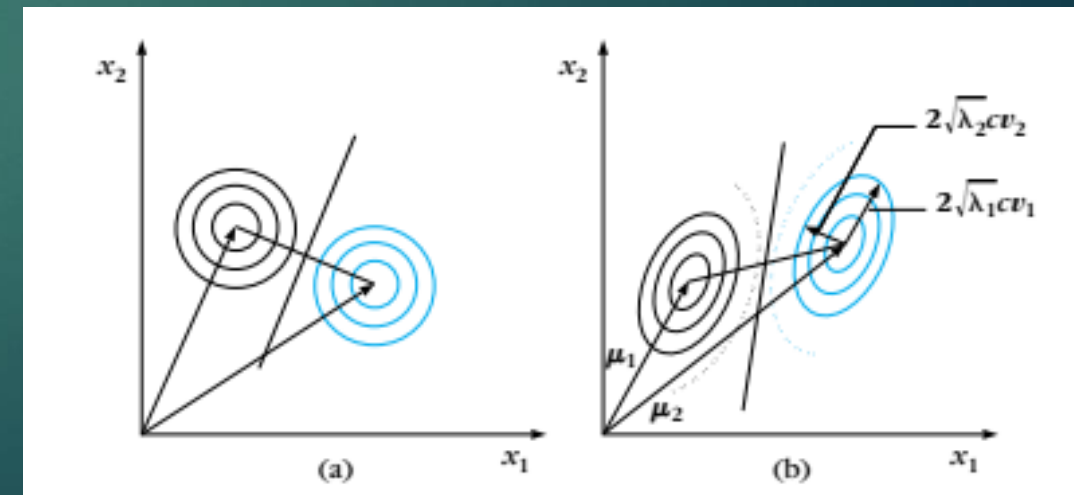
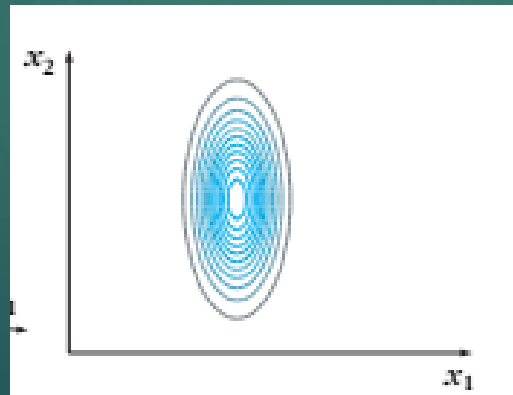
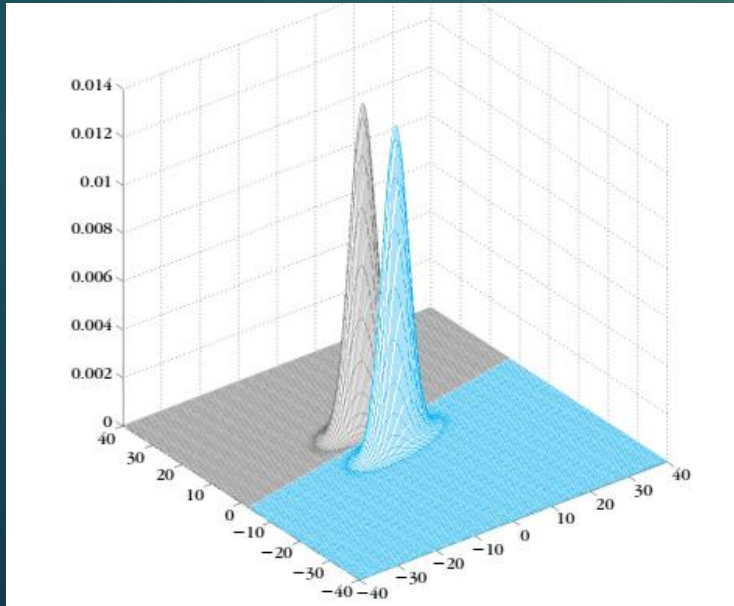
Multivariate Normal Distribution- MAP Classifier

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- Case 2 $\Sigma_i = \Sigma$ (equal, general)

$$-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_i| + \ln(p(\omega_i)) = \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) + \ln P(\omega_i)$$

- Boundaries rotated by $\boldsymbol{\Sigma}^{-1}$

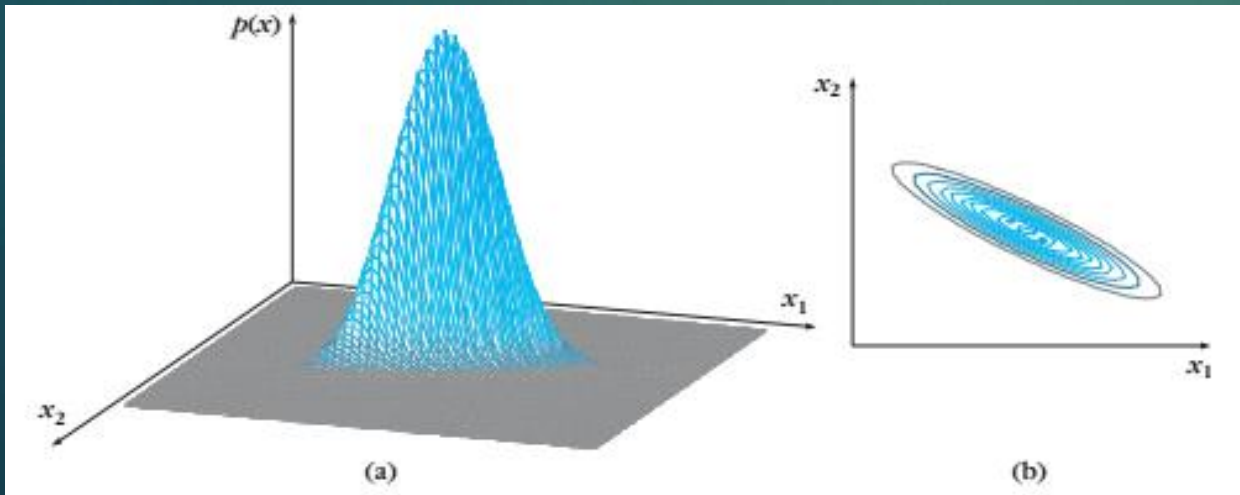


Multivariate Normal Distribution- MAP Classifier

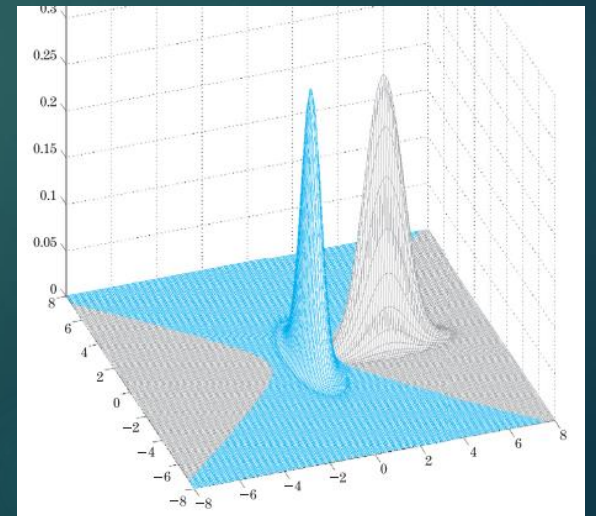
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- Case 3 Full Σ_i (class-specific)

$$-\frac{1}{2} (x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_i| + \ln(p(\omega_i)) = x^T W_i x + w_i^T x + w_{i0}$$



- Boundaries are hyperquadrics (conic section)



Arbitrarily Distributed Features

SECTION 3.3.6

WHEN A SINGLE GAUSSIAN IS NOT ENOUGH · GAUSSIAN MIXTURE MODELS

The Problem: When One Gaussian Fails

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- A single Gaussian may be insufficient when the feature distribution is skewed or has a multimodal shape — i.e., multiple clusters within the same class.

✓ Standard Case — Works Well

- Each class follows a single unimodal distribution
- Parameters μ_j and Σ_j can be estimated from training data
- Decision boundaries are conic sections (as in General case for Σ)

✗ Problem Case — Gaussian Breaks Down

- Feature distribution is skewed or multimodal
- Hidden sub-classes or latent factors within a class
- Single Gaussian forces a poor, averaged fit to the data

Motivating Example — The Iris Dataset

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Classification Scenario

Task A — Species within Iris

Classify Iris setosa, versicolor, virginica. Each class has its own Gaussian → standard Bayesian classifier works.

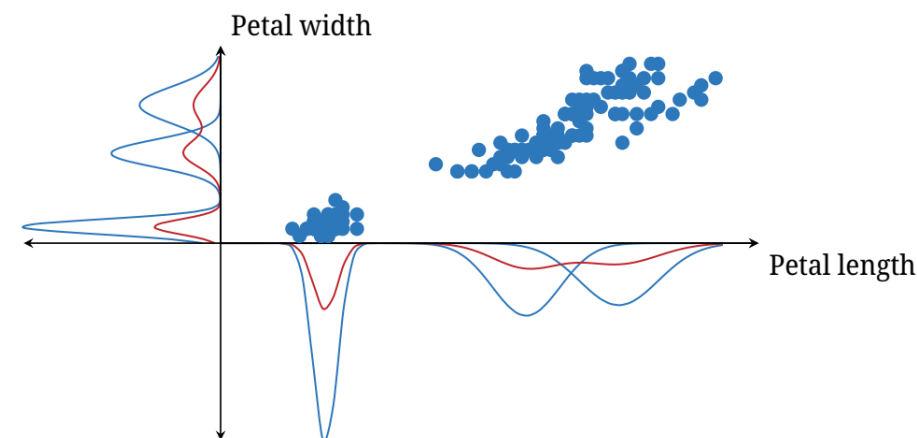
Task B — Genus level

Distinguish Iris vs Gladiolus. Now a single class must cover multiple Iris sub-species. One Gaussian is too rigid.

The Core Difficulty

Without knowledge of sub-species, we cannot identify the latent factors causing multimodality. A single Gaussian will produce a poor, averaged density.

GMM on Iris Data



Blue curves: individual Gaussian per sub-component.

Red curve: their convex combination (the GMM).

Solution: use K=3 Gaussian components to fit the Iris class

Gaussian Mixtures

SECTION 3.4

Gaussian Mixture Model (GMM)

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- A random variable $\mathbf{x} \in \mathbb{R}^d$ is a Gaussian mixture with K components if its density is:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

With constraints:

$$0 \leq \pi_k \leq 1 \quad \text{and} \quad \sum_{k=1}^K \pi_k = 1$$

K	Number of mixture components (Gaussians)
π_k	Mixing weight — how much component k contributes (prior probability)
$\boldsymbol{\mu}_k$	Mean vector of the k -th Gaussian component
$\boldsymbol{\Sigma}_k$	Covariance matrix of the k -th Gaussian component
$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$	Multivariate Gaussian density with parameters $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$

Universal Approximation Property

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- Approximation Theorem

- Let “f” be any density with a finite number of discontinuities on each compact subset of its support.
- There exists a sequence of Gaussian mixtures $f_n = \sum_{k=1}^{K_n} \pi_{nk} \mathcal{N}(m; \mu_{nk}, \Sigma_{nk})$ that converges uniformly to “f”, except at discontinuity points.



Any Distribution Approximable

With enough components ($K \rightarrow \infty$), a GMM can model any well-behaved continuous density — unimodal, multimodal, skewed, heavy-tailed.



No Closed-Form Optimum

There is no closed-form solution to find the optimal parameters. Iterative algorithms like EM are required (Section 4.5).



Unknown Number of Components

No general rule exists to determine K for a given approximation error. K is typically a hyperparameter chosen by model selection.



Components May Change

Each f_n may have entirely different components — μ_{nk} and Σ_{nk} are not necessarily a superset of the previous mixture.

Bayesian Interpretation of Mixing Weights π_k

- The mixing coefficients π_k can be interpreted as a priori probabilities $\pi_k = P(k)$ over the “K” components.
- This enables a Bayesian perspective on Gaussian mixtures.
- Posterior Probability of Component “k”

$$p(k|x) = \frac{P(k) \cdot p(x|k)}{\sum_i P(i) \cdot P(x|i)} = \frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{\sum_i \pi_i \mathcal{N}(x; \mu_i, \Sigma_i)}$$

Prior	$P(k) = \pi_k$	Likelihood	$p(x k)$	Posterior	$p(k x)$
Probability of component “k” before observing “x”. Set by the mixing weights.		Probability of observing “x” given it came from component “k”. This is $\mathcal{N}(x; \mu_k, \Sigma_k)$.		Probability that m was generated by component “k”, after seeing “x”. Used in EM.	

GMM vs Single Gaussian — Key Differences

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Single Gaussian		VS	Gaussian Mixture Model
Shape	Unimodal (one peak)		Multimodal (K peaks possible)
Parameters	μ, Σ (2 params)		π_k, μ_k, Σ_k for $k=1\dots K$
Flexibility	Low — fixed ellipsoidal shape		High — arbitrary complex shapes
Boundaries	Conic sections (hyperquadrics)		Arbitrarily complex boundaries
When to use	Unimodal, approximately Gaussian data		Skewed, multimodal, clustered data
Risk	Poor fit if data is multimodal		Risk of overfitting with large K

How a GMM Works

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- A GMM is a convex combination of Gaussians — each component explains part of the data, weighted by π_k .

1

Choose K

Decide how many Gaussian components to use.
Each component will capture one 'sub-cluster' of the data.

2

Assign Weights π_k

Each component k has a mixing weight π_k (prior probability). They must sum to 1.

3

Define Each Gaussian

Fit parameters μ_k and Σ_k for each component — mean and covariance of that sub-cluster.

4

Combine

The final density is $p(x) = \sum \pi_k \mathcal{N}(x; \mu_k, \Sigma_k)$ — a weighted sum that can represent any shape.

